

ULAB Setu (সেতু)

"Where ULAB students meet the alumni who came before them."

Target Institution: University of Liberal Arts
Bangladesh (ULAB)

Database & ORM: Prisma ORM (SQLite /
PostgreSQL)

Tech Stack: Next.js 16, React 19, TypeScript 5,
Tailwind v4

Report Type: Architecture, Features & Project
Summary

1. Executive Summary

ULAB Setu (Bengali for *Bridge*) is a modern, high-performance web platform and Progressive Web App (PWA) built specifically to bridge the divide between undergraduate students, graduated alumni, and university faculty. By providing a secure, verified institutional network, Setu eliminates cold outreach friction and connects students directly to career opportunities, employee referrals, academic mentorship, and campus announcements.

Verified Campus Network

Institutional authentication with strict student domain gating (`@uLab.edu.bd`) and admin verification for alumni & teachers.

3-Tier Job Aggregator

Internal alumni & faculty job circulars (with referral flags), Bangladesh national jobs via Careerjet/Bdjobs, and global remote jobs via Himalayas.

Frictionless Direct Messaging

Instant 1-on-1 peer and mentor messaging without waiting for connection handshake approvals.

Installable PWA Experience

Fully installable on Android and iOS devices, with service worker offline caching and native app behavior.

2. System Architecture & Technology Stack

The project leverages the latest generation of web technologies, adopting the cutting-edge Next.js 16 App Router, React 19, and Tailwind CSS v4 design tokens.

Layer	Technology	Key Implementation Purpose
Frontend & UI	Next.js 16 (App Router), React 19, TypeScript 5	Server-side rendering (SSR), streaming, and type-safe server actions.
Styling & Design	Tailwind CSS v4, Motion (Framer Motion)	High-performance CSS tokens, smooth stagger transitions, and dark/light mode.
3D Graphics	Three.js WebGL Particle Engine	Interactive 3D particle morphing (globe → cap → bridge → briefcase).
Database & ORM	Prisma ORM 6 (SQLite / PostgreSQL)	Type-safe schema, relational migrations, and automated seeding.
Security & Auth	Jose (JWT), BcryptJS, Zod	HTTP-only cookie sessions, cryptographic validation, and input sanitization.
Access Control	Custom Next.js Proxy (src/proxy.ts)	Strict route protection for /student , /alumni , /teacher , /admin .
External APIs	Careerjet API & Himalayas API	Live job listings for Bangladesh and international remote tech positions.
Testing	Playwright-Core & Headless Chrome	Full E2E smoke tests simulating multi-role logins, jobs, chats, and mobile viewport.

3. User Roles & Access Control Matrix

The platform incorporates four specialized user roles, with distinct interfaces, capabilities, and approval flows:

Role	Onboarding & Verification	Primary Permissions & Capabilities
Student	Instant verification restricted to <code>@uLab.edu.bd</code> institutional email.	Browse 3-tier jobs, apply with note once profile is filled, search people directory, message mentors, view notice board.
Alumni	Requires Student ID & Graduation Year; reviewed by Alumni Office.	Post job openings with internal employee referral badges, review/shortlist student applicants, post notices/scholarships, chat.
Teacher	Requires department & academic rank selection; reviewed by Admin.	Post Teaching Assistant (TA) & Research Assistant (RA) jobs, publish departmental notices, seminars, and hackathon events.
Admin (Alumni Office)	Superuser authority (preset administrator account).	Review pending alumni & teacher registrations with one-click approve/reject actions to preserve network integrity.

4. Core Functional Modules

4.1. Three-Tier Job Portal

- **Alumni & Teacher Openings:** Native job postings from verified university graduates and professors. Alumni can mark jobs with a prominent *"Employee Referral"* badge to fast-track student resumes.
- **Bangladesh National Openings:** Powered by the Careerjet API (`locale_code=en_BD`), aggregating opportunities from Bdjobs and major national companies.
- **Global Remote Roles:** Direct integration with the Himalayas API, providing students access to international remote internships and junior engineering positions.
- **External Job Portals Quick-Links:** Direct links to Bdjobs, Teletalk All Jobs (government circulars), LinkedIn Bangladesh, Skill.jobs, Chakri, etc.

4.2. Notice Board & Campus Announcements

Teachers and alumni can broadcast critical academic and career information across five categories:

- 📢 **Notice:** University announcements and career guidelines.
- 📅 **Event:** Seminars, guest lectures, hackathons, and alumni meetups.
- 🎓 **Scholarship:** Graduate funding, international fellowship alerts, and study grants.
- 🔬 **Research:** Faculty research collaborations and call for undergraduate research assistants.
- 💡 **Opportunity:** Competitions, training camps, and leadership summits.

4.3. Privacy & Granular Contact Visibility

Users can independently control whether their **Phone Number**, **WhatsApp**, and **Facebook** profiles are Public or Private. Hidden details are filtered out server-side, preventing unconsented contact while maintaining easy communication.

4.4. Visual Excellence & Motion Design

- **3D Three.js Particle Morphing:** As students scroll the landing page, an interactive particle cloud morphs through four milestones: a rotating globe → a graduation cap → a connecting bridge → a career briefcase.
- **Physics-based Corner Balls:** Interactive spring-physics balls that animate across the viewport upon interaction.
- **Page-Pull Transition & Dark Mode:** Smooth route transitions and persistent theme toggling.

5. Database Schema Overview (Prisma ORM)

```
model User {
  id          String          @id @default(cuid())
  name        String
  email       String          @unique
  role        String          // STUDENT | ALUMNI | TEACHER | ADMIN
  status      String          @default("APPROVED") // PENDING | APPROVED | REJECTED
  department  String?
  studentId   String?
  graduationYear String?
  company     String?
  designation  String?
  phonePublic Boolean          @default(false)
  whatsappPublic Boolean       @default(false)
  facebookPublic Boolean       @default(false)
  jobs        Job[]
  applications Application[]
  posts       Post[]
  messages    Message[]
}

model Job {
  id          String          @id @default(cuid())
  title       String
  company     String
  location    String
  type        String          // Full-time | Part-time | Internship | Contract
  referral    Boolean          @default(false)
  active      Boolean          @default(true)
  postedById  String
  postedBy    User            @relation(fields: [postedById], references: [id])
  applications Application[]
}

model Application {
  id          String          @id @default(cuid())
  note        String?
  status      String          @default("APPLIED") // APPLIED | SHORTLISTED | REJECTED
  jobId       String
  studentId   String
  job         Job             @relation(...)
  student     User            @relation(...)
  @@unique([jobId, studentId])
}
```

6. Execution & Deployment Guide

Zero-Config Local Run: Setu includes a built-in SQLite database and comprehensive mock seeder for immediate local testing.

```
# 1. Install dependencies
pnpm install

# 2. Environment setup
cp .env.example .env
# Set AUTH_SECRET (e.g. openssl rand -hex 32)
# Set CAREERJET_API_KEY (optional, fallback to mock data)

# 3. Database initialization & seed
pnpm db:push
pnpm db:seed

# 4. Start development server
pnpm dev
# Server ready at http://localhost:3000
```

Pre-Configured Demo Accounts (Password: `password123`):

- **Student:** `student@ulab.edu.bd`
- **Alumni (Approved):** `alumni1@example.com` to `alumni6@example.com`
- **Alumni (Pending Approval):** `pending.alumni@example.com`
- **Teacher (Approved):** `teacher@ulab.edu.bd`
- **Admin (Alumni Office):** `admin@ulab.edu.bd`

7. Conclusion

ULAB Setu represents an exceptional blend of modern full-stack web engineering, rigorous access control, and dynamic visual storytelling. It delivers a turnkey digital bridge empowering students with mentorship and employment opportunities, while allowing alumni and teachers to give back meaningfully to the university community.